Chapter 6

External Memory

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Magnetic Disk

• القرص هو طبق دائري شيدت من المواد غير المتغيرة ، وتسمى الركيزة ، والمغلفة مع مادة قابلة للتغذية

- A disk is a circular *platter* constructed of nonmagnetic material, called the *substrate,* coated with a magnetizable material
  - Traditionally the substrate has been an aluminium or aluminium alloy material
  - Recently glass substrates have been introduced
- Benefits of the glass substrate:
  - Improvement in the uniformity of the magnetic film surface to increase disk reliability
  - A significant reduction in overall surface defects to help reduce read-write errors
  - Ability to support lower fly heights
  - Better stiffness to reduce disk dynamics
  - Greater ability to withstand shock and damage



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Magnetic Read and Write Mechanisms

كيف رح اتعامل معها جاي بدور الراين

Data are recorded on and later retrieved from the disk via a conducting coil named the *head*

- In many systems there are two heads, a read head and a write head
- During a read or write operation the head is stationary while the platter rotates beneath it

في اقطعة بيكون هدا 2 هيد بعض راسين واحد للرايت وواحد للريد

وفي اقطعة بس راس واحد بيكون read write

The write mechanism exploits the fact that electricity flowing through a coil produces a magnetic field

تعتمد على التيار الكهربائي وتحويله الى مجال مغناطيسي

Electric pulses are sent to the write head and the resulting magnetic patterns are recorded on the surface below, with different patterns for positive and negative currents

نقل البيانات في الرايت من طريق النبضات بالتيار الكهربائي

The write head itself is made of easily magnetizable material and is in the shape of a rectangular doughnut with a gap along one side and a few turns of conducting wire along the opposite side

كيف تتم عمليات الراين؟

An electric current in the wire induces a magnetic field across the gap, which in turn magnetizes a small area of the recording medium

تتراكم تيار كهربائي في الأسلاك حقل مغناطيسي عبر الفجوة ، والذي بدوره يشترك في مساحة صغيرة من وسيلة التسجيل . تخزن على شكل باينري 0,1

Reversing the direction of the current reverses the direction of the magnetization on the recording medium

* الراين لو حنط البيانات بما يسمى التيار الكهربائي عبر مايلات خاصة
* بيكون الهيد رايت راسين بيكون بيرو متايل للتحدد

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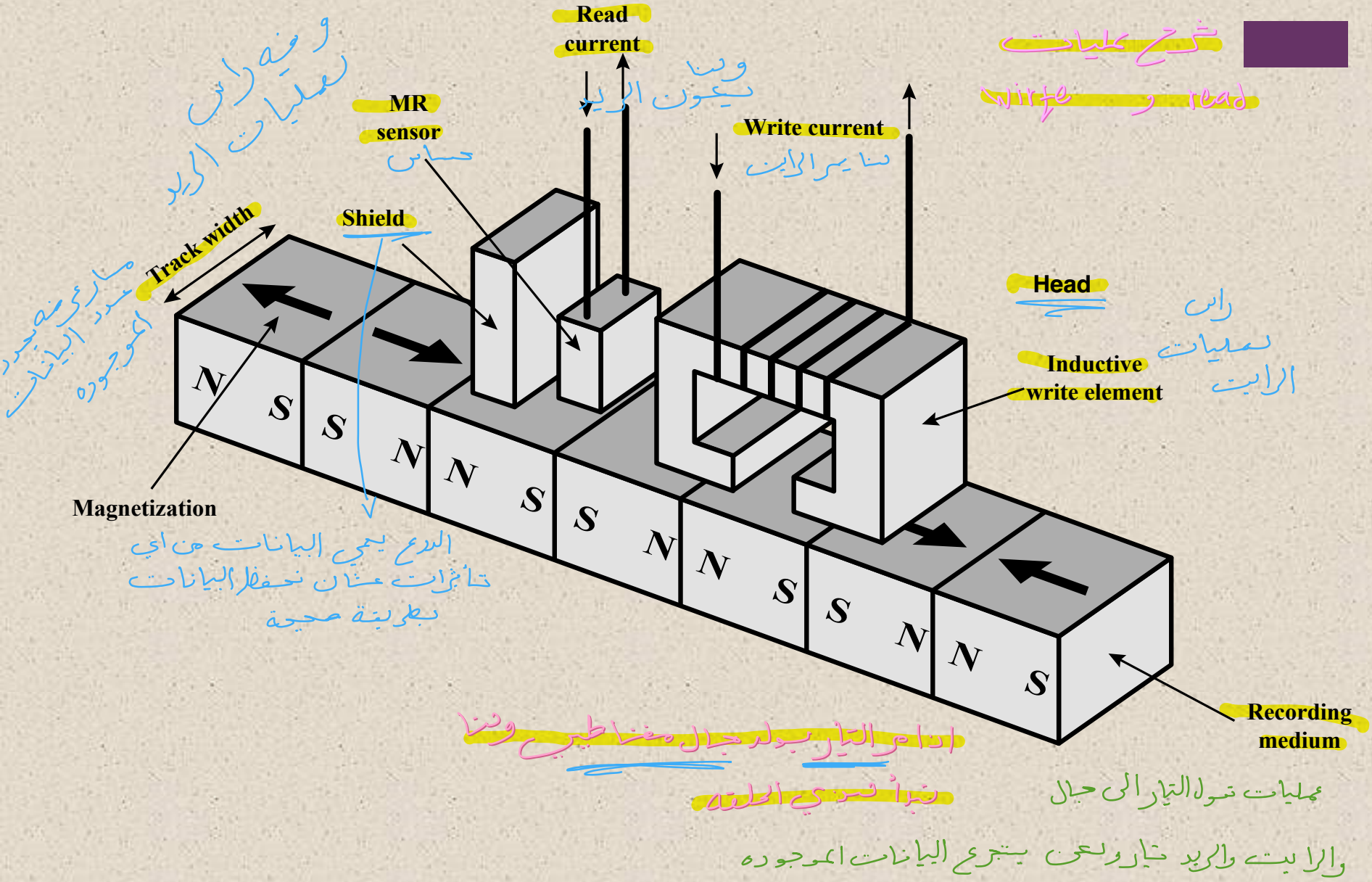


Figure 6.1 Inductive Write/Magnetoresistive Read Head

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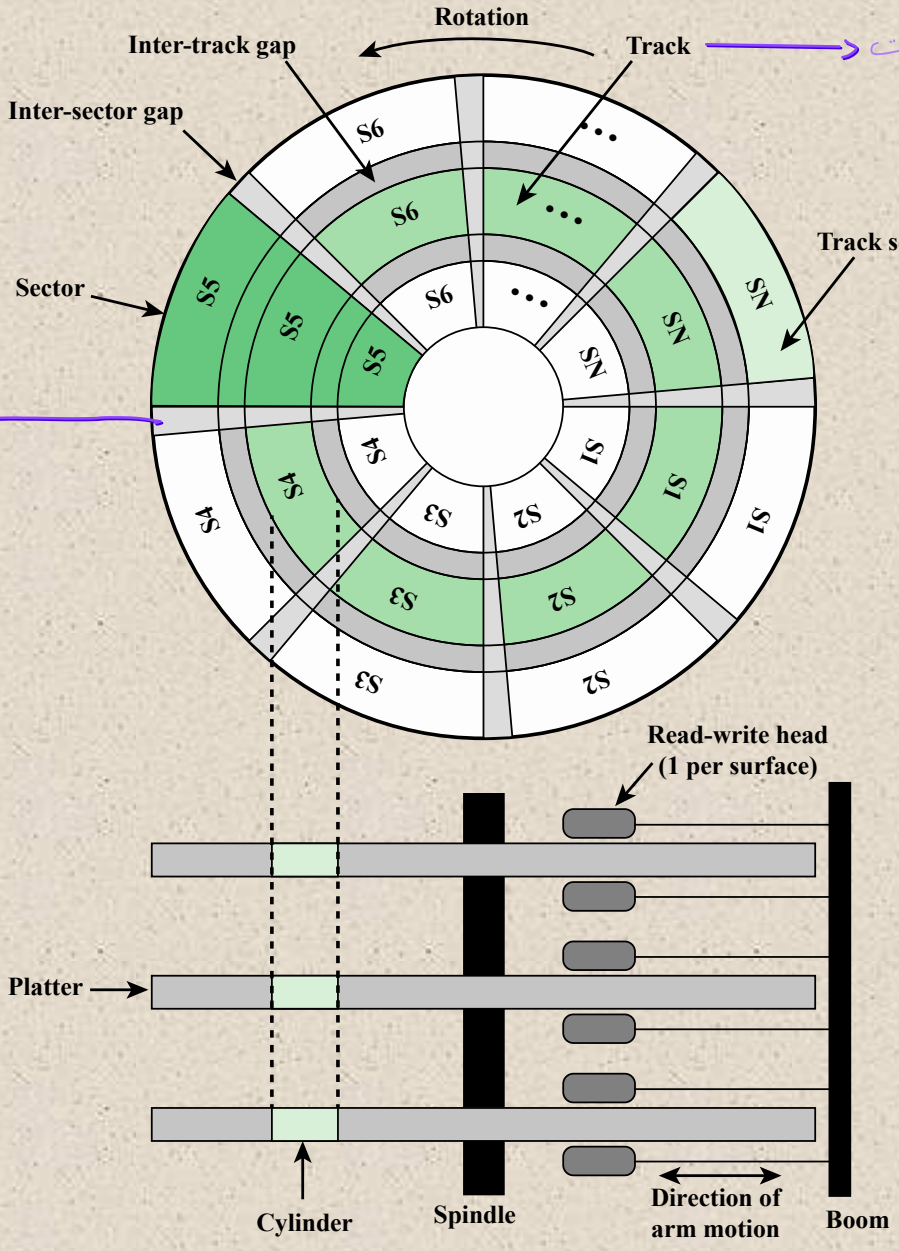


Figure 6.2 Disk Data Layout

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Characteristics

- **Fixed-head disk**
  - One read-write head per track
  - Heads are mounted on a fixed ridged arm that extends across all tracks
- **Movable-head disk**
  - One read-write head
  - Head is mounted on an arm
  - The arm can be extended or retracted
- **Non-removable disk**
  - Permanently mounted in the disk drive
  - The hard disk in a personal computer is a non-removable disk
- **Removable disk**
  - Can be removed and replaced with another disk
  - Advantages:
    - Unlimited amounts of data are available with a limited number of disk systems
    - A disk may be moved from one computer system to another
  - Floppy disks and ZIP cartridge disks are examples of removable disks
- **Double sided disk**
  - Magnetizable coating is applied to both sides of the platter



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Disk Classification

The head mechanism provides a classification of disks into three types

تقسيم الديلد راهم جزء منه الـ head لانه يسوي الـ read و write

- The head must generate or sense an electromagnetic field of sufficient magnitude to write and read properly
- The narrower the head, the closer it must be to the platter surface to function
  - A narrower head means narrower tracks and therefore greater data density
- The closer the head is to the disk the greater the risk of error from impurities or imperfections

Winchester Heads

- Used in sealed drive assemblies that are almost free of contaminants
- Designed to operate closer to the disk's surface than conventional rigid disk heads, thus allowing greater data density
- Is actually an aerodynamic foil that rests lightly on the platter's surface when the disk is motionless
  - The air pressure generated by a spinning disk is enough to make the foil rise above the surface

خلص الكلام ذا كله من الدكتور

*الهيد مسؤول عن كل عمليات الريد و الرايت وينشى التيار عشان ينقل البيانات

*كل ماكانت المساحه اضيق يكون اقرب من الديسك ويصير دقه في الريد و الرايت

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Disk Performance Parameters

اعمار الدسك

- When the disk drive is operating the disk is rotating at constant speed
  الديسك عباره عن روتيشين دوران بسرعه ثابته
- To read or write the head must be positioned at the desired track and at the beginning of the desired sector on the track
  - Track selection involves moving the head in a movable-head system or electronically selecting one head on a fixed-head system
  - Once the track is selected, the disk controller waits until the appropriate sector rotates to line up with the head

لو الهيد يتحرك هو يسمح لي احرك كل شي
لو كان الهيد فيكسد ثابت يتم اختيار التراك الي ثابا عليه الكترونيا

- Seek time
  - On a movable–head system, the time it takes to position the head at the track
    الوقت اللازم للهيد عشان يحدد المسار الصحيح اي تراك مفروض يمشي عليه
- Rotational delay *(rotational latency)*
  - The time it takes for the beginning of the sector to reach the head
    الوقت الي يمكن يستغرقه لتحريك القرص كامل عشان اوصل او يحدد اي سيكتور مطلوب

*التراك المسار كامل كله وكل مسار فيه سيكتور

- Access time
  - The sum of the seek time and the rotational delay
  - The time it takes to get into position to read or write
    وقت الوصول لما يكونو كلهم السيك تايم و الرولاتيشن الي فوق الغريبه ذي خاص في
- Transfer time
  كل ريد ورايد يحدد التراك الخاص او السيكتور ويتم حفظ البيانات
  - Once the head is in position, the read or write operation is then performed as the sector moves under the head
  - This is the data transfer portion of the operation
    عمليات الريد يحتاج الى عمليات ترانسيفرر تايم الي هو بعد ما يخلص الاكسس تايم



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RAID

Redundant Array of Independent Disks

- Consists of 7 levels
- Levels do not imply a hierarchical relationship but designate different design architectures that share three common characteristics:
  1) Set of physical disk drives viewed by the operating system as a single logical drive
  2) Data are distributed across the physical drives of an array in a scheme known as striping
  3) Redundant disk capacity is used to store parity information, which guarantees data recoverability in case of a disk failure

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RAID Level 0

- Addresses the issues of request patterns of the host system and layout of the data
- Impact of redundancy does not interfere with analysis

Raid 0

RAID 0 for High Data Transfer Capacity

- For applications to experience a high transfer rate two requirements must be met:

زياده رفع عدد البيانات اعلى معدل نقل للبيانات

الهدف الاساسي معدل عالي من البيانات

1. A high transfer capacity must exist along the entire path between host memory and the individual disk drives

نقل البيانات تكون بين الميموري و الديسك نفسه

2. The application must make I/O requests that drive the disk array efficiently

البيانات على كذا قرص وكل قرص له نوع معين من البيانات يخلي لي الريد والرايت بصوره متوازيه ومن مكانين مختلفين

الميزه الاساسيه مافيها اي نسخه للبيانات ادائها عالي ورح يؤدي الى عدم وجود نسخ احتياطي لان البيانات تتأثر في زمن الوصول للسيكتور او لزمن الوصول للتراك

RAID 0 for High I/O Request Rate

- For an individual I/O request for a small amount of data the I/O time is dominated by the seek time and rotational latency
- A disk array can provide high I/O execution rates by balancing the I/O load across multiple disks
- If the strip size is relatively large multiple waiting I/O requests can be handled in parallel, reducing the queuing time for each request

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RAID Level 1

*وجود عمليه الكوبي من جميع البيانات
*نسخ البيانات بقرص مختلف ما تصحح لي الاخطاء وولا شي بس تسوي لها نسخ

*القرص الاول البيانات الاصليه اي تعديل واضافه للقرص الاخر كمان لو
صار فشل تروح للقرص الثاني او العكس وهنا رح يتأكد من الغلط وصححه

Raid 1

Characteristics

- Differs from RAID levels 2 through 6 in the way in which redundancy is achieved
- Redundancy is achieved by the simple expedient of duplicating all the data
- Data striping is used but each logical strip is mapped to two separate physical disks so that every disk in the array has a mirror disk that contains the same data
- RAID 1 can also be implemented without data striping, although this is less common

Positive Aspects

- A read request can be serviced by either of the two disks that contains the requested data
- There is no "write penalty"
- Recovery from a failure is simple, when a drive fails the data can be accessed from the second drive
- Provides real-time copy of all data
- Can achieve high I/O request rates if the bulk of the requests are reads
- Principal disadvantage is the cost

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RAID Level 2

*استخدام الكارولیل آكسید تقنیه الوصول عملیه الرید و الرایت متزامنه

*تعتمد على تصحيح الاخطاء

*يصحح الاخطاء عن طريق acc ويتم تخزين البينات كمان على acc و الهامينق كود في تصحيح اي اخطاء

R 2
a
i
d

Characteristics

- Makes use of a parallel access technique
- In a parallel access array all member disks participate in the execution of every I/O request
- Spindles of the individual drives are synchronized so that each disk head is in the same position on each disk at any given time
- Data striping is used
  - Strips are very small, often as small as a single byte or word

Performance

- An error-correcting code is calculated across corresponding bits on each data disk and the bits of the code are stored in the corresponding bit positions on multiple parity disks
- Typically a Hamming code is used, which is able to correct single-bit errors and detect double-bit errors
- The number of redundant disks is proportional to the log of the number of data disks
- Would only be an effective choice in an environment in which many disk errors occur

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RAID Level 3

*بوزه حسان للبيانات
داخله بالرني

الفكره الاساسيه حمايه البيانات
لو صار اي مشكله بأي ديسك يبدلونه بقرص آخر

Raid

3

Redundancy

- Requires only a single redundant disk, no matter how large the disk array
- Employs parallel access, with data distributed in small strips
- Instead of an error correcting code, a simple parity bit is computed for the set of individual bits in the same position on all of the data disks
- Can achieve very high data transfer rates

Performance

- In the event of a drive failure, the parity drive is accessed and data is reconstructed from the remaining devices
- Once the failed drive is replaced, the missing data can be restored on the new drive and operation resumed
- In the event of a disk failure, all of the data are still available in what is referred to as *reduced mode*
- Return to full operation requires that the failed disk be replaced and the entire contents of the failed disk be regenerated on the new disk
- In a transaction-oriented environment performance suffers

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RAID Level 4

Characteristics

- Makes use of an independent access technique
  - In an independent access array, each member disk operates independently so that separate I/O requests can be satisfied in parallel
- Data striping is used
  - Strips are relatively large
- To calculate the new parity the array management software must read the old user strip and the old parity strip

Performance

- Involves a write penalty when an I/O write request of small size is performed
- Each time a write occurs the array management software must update not only the user data but also the corresponding parity bits
- Thus each strip write involves two reads and two writes

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RAID Level 5

Characteristics

- Organized in a similar fashion to RAID 4
- Difference is distribution of the parity strips across all disks
- A typical allocation is a round-robin scheme
- The distribution of parity strips across all drives avoids the potential I/O bottleneck found in RAID 4

RAID Level 6

Characteristics

- Two different parity calculations are carried out and stored in separate blocks on different disks
- Advantage is that it provides extremely high data availability
- Three disks would have to fail within the mean time to repair (MTTR) interval to cause data to be lost
- Incurs a substantial write penalty because each write affects two parity blocks

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Level	Advantages	Disadvantages	Applications
0	I/O performance is greatly improved by spreading the I/O load across many channels and drives No parity calculation overhead is involved Very simple design Easy to implement	The failure of just one drive will result in all data in an array being lost	Video production and Editing Image editing Pre-press applications Any application requiring high bandwidth
1	100% redundancy of data means no rebuild is necessary in case of a disk failure, just a copy to the replacement disk Under certain circumstances, RAID 1 can sustain multiple simultaneous drive failures Simplest RAID storage subsystem design	Highest disk overhead of all RAID types (100%) - inefficient	Accounting Payroll Financial Any application requiring very high availability
2	Extremely high data transfer rates possible The higher the data transfer rate required, the better the ratio of data disks to ECC disks Relatively simple controller design compared to RAID levels 3,4 & 5	Very high ratio of ECC disks to data disks with smaller word sizes - inefficient Entry level cost very high - requires very high transfer rate requirement to justify	No commercial implementations exist / not commercially viable

Table 6.4

RAID Comparison (page 1 of 2)

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3	Very high read data transfer rate Very high write data transfer rate Disk failure has an insignificant impact on throughput Low ratio of ECC (parity) disks to data disks means high efficiency	Transaction rate equal to that of a single disk drive at best (if spindles are synchronized) Controller design is fairly complex	Video production and live streaming Image editing Video editing Prepress applications Any application requiring high throughput
4	Very high Read data transaction rate Low ratio of ECC (parity) disks to data disks means high efficiency	Quite complex controller design Worst write transaction rate and Write aggregate transfer rate Difficult and inefficient data rebuild in the event of disk failure	No commercial implementations exist / not commercially viable
5	Highest Read data transaction rate Low ratio of ECC (parity) disks to data disks means high efficiency Good aggregate transfer rate	Most complex controller design Difficult to rebuild in the event of a disk failure (as compared to RAID level 1)	File and application servers Database servers Web, e-mail, and news servers Intranet servers Most versatile RAID level
6	Provides for an extremely high data fault tolerance and can sustain multiple simultaneous drive failures	More complex controller design Controller overhead to compute parity addresses is extremely high	Perfect solution for mission critical applications

Table 6.4

RAID Comparison (page 2 of 2)

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SSD Compared to HDD

- SSDs have the following advantages over HDDs:
  - High-performance input/output operations per second (IOPS)
  - Durability
  - Longer lifespan
  - Lower power consumption
  - Quieter and cooler running capabilities
  - Lower access times and latency rates

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Practical Issues

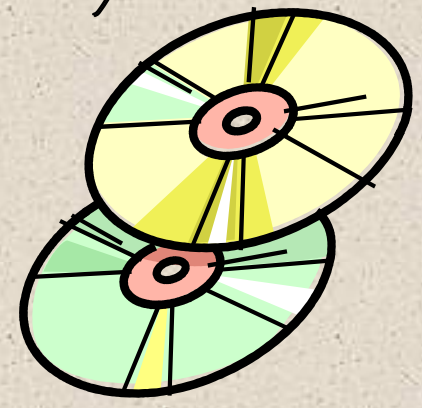
There are two practical issues peculiar to SSDs that are not faced by HDDs:

- SDD performance has a tendency to slow down as the device is used
  - The entire block must be read from the flash memory and placed in a RAM buffer
  - Before the block can be written back to flash memory, the entire block of flash memory must be erased
  - The entire block from the buffer is now written back to the flash memory
- Flash memory becomes unusable after a certain number of writes
  - Techniques for prolonging life:
    - Front-ending the flash with a cache to delay and group write operations
    - Using wear-leveling algorithms that evenly distribute writes across block of cells
    - Bad-block management techniques
- Most flash devices estimate their own remaining lifetimes so systems can anticipate failure and take preemptive action

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CD-ROM

- CD-ROM is appropriate for the distribution of large amounts of data to a large number of users
- Because the expense of the initial writing process it is not appropriate for individualized applications
- The CD-ROM has two advantages:
  - The optical disk together with the information stored on it can be mass replicated inexpensively
  - The optical disk is removable, allowing the disk itself to be used for archival storage
- The CD-ROM disadvantages:
  - It is read-only and cannot be updated
  - It has an access time much longer than that of a magnetic disk drive



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CD Recordable (CD-R)

بس اقدر اكتب مره وحده

- Write-once read-many
- Accommodates applications in which only one or a small number of copies of a set of data is needed
- Disk is prepared in such a way that it can be subsequently written once with a laser beam of modest-intensity
- Medium includes a dye layer which is used to change reflectivity and is activated by a high-intensity laser
- Provides a permanent record of large volumes of user data

CD Rewritable (CD-RW)

اقدر اكتب اكثر من مره

- Can be repeatedly written and overwritten
- Phase change disk uses a material that has two significantly different reflectivities in two different phase states
- Amorphous state
  - Molecules exhibit a random orientation that reflects light poorly
- Crystalline state
  - Has a smooth surface that reflects light well
- A beam of laser light can change the material from one phase to the other
- Disadvantage is that the material eventually and permanently loses its desirable properties
- Advantage is that it can be rewritten

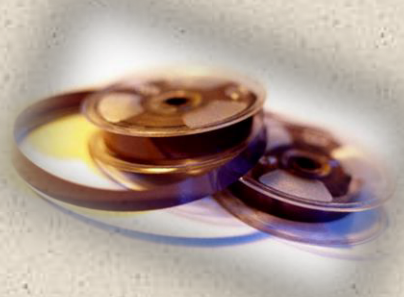
يعتمد على الـ phase change تغير الحالات

اخزن فيها اكثر من مره ممكنه

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Magnetic Tape

- Tape systems use the same reading and recording techniques as disk systems
- Medium is flexible polyester tape coated with magnetizable material
- Coating may consist of particles of pure metal in special binders or vapor-plated metal films
- Data on the tape are structured as a number of parallel tracks running lengthwise
- Serial recording
  - Data are laid out as a sequence of bits along each track
- Data are read and written in contiguous blocks called *physical records*
- Blocks on the tape are separated by gaps referred to as *inter-record gaps*



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